

Thyroid Storm



Section I: Scenario Demographics

Scenario Title:	Thyroid Storm
Date of Development:	(05/03/2015)
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY $\geq$ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Cheryl ffrench
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To review the presentation and management of thyrotoxicosis.
CRM Objectives:	1) Avoid fixation error/anchoring by maintaining a broad differential diagnosis 2) Use leadership style that encourages team members to offer suggestions for management in a complex patient
Medical Objectives:	1) Recognize and appropriately manage a patient with congestive heart failure in the context of thyroid storm. 2) Initiate appropriate thyrotoxicosis treatment with assistance from consultants 3) Calmly discuss patient's status at bedside with husband, including need for intubation

Case Summary: Brief Summary of Case Progression and Major Events

A 31 year-old-female presents by EMS with altered LOC and fever due to thyroid storm precipitated by recent parturition. The patient is tachycardic and hypoxic on arrival. Her level of consciousness will continue to deteriorate despite IV fluids and antibiotics and will require intubation. The husband will be at the bedside, and the team will need to discuss the need for intubation with him. After intubation, lab results will come back indicating possible thyrotoxicosis. The patient's rhythm will change to atrial fibrillation at this time. The team will be expected to manage the thyroid storm in consultation with Endocrinology and ICU.

References

Choudhury RP & MacDermot J. (1998). Heart failure in thyrotoxicosis, an approach to management. *British journal of clinical pharmacology*. <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1873689/>



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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You are working the evening shift at a tertiary care hospital. A 31-year-old female two weeks postpartum is brought in by EMS accompanied by her husband. He is concerned because she is delirious and somewhat difficult to rouse.			
B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☒ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
C. Required Monitors			
☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☐ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☐ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
E. Moulage			
Female mannequin wearing jogging pants and a t-shirt.			
F. Approximate Timing			
Set-Up:	3 min	Scenario:	10 min
		Debriefing:	15 min

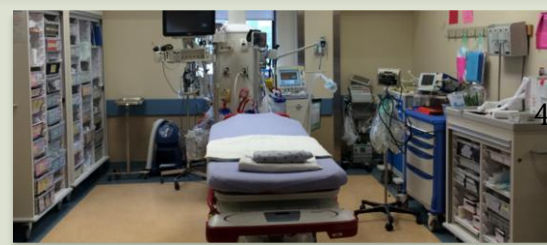
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Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Sara Lindy		Age: 31		Weight: 70kg	
Gender: ☐ M ☒ F		Code Status: Full Code			
Chief Complaint: Delirium, altered LOC and fever					
History of Presenting Illness: Patient is two weeks post partum. This morning the patient was found to be confused and difficult to rouse confused so the husband called 911. He indicates that she's been very anxious since delivering her first baby (2 weeks ago). Symptoms include two days of nausea, vomiting and watery-diarrhea associated with generalized abdominal pain. Last emesis 3 hours prior to presentation. Hasn't been able to breastfeed baby because has been too anxious. No sick contacts. No recent travel or antibiotics.					
Past Medical History:	G1P1 (2 weeks post partum)		Medications:	Alprazolam	
	Anxiety				
Allergies: penicillin					
Social History: Patient lives with husband and newborn child at home.					
Family History: Nil					
Review of Systems:	CNS:	Delirious, decreased LOC. When roused (sternal rub – eyes half-way open) is confused and disorientated/delirious (will mumble - "I have to feed the cat, I need to wash the car; the plane is about to leave I have to go").			
	HEENT:	Normal			
	CVS:	Normal			
	RESP:	Normal			
	GI:	Nauseated. Last emesis 3 hours before presentation – vomiting 2-3 times for two days. Daily episodes of non-bloody diarrhea.			
	GU:	Normal			
	MSK:	Normal	Psych:	Anxious	
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 145 /min	BP: 150/80	RR: 32	O ₂ SAT: 89% 5L NP		
Rhythm: regular	T: 40.0 °C	Glucose: 6.2 mmol/L	GCS: 12 (E3 V5 M4)		
General Status: looks unwell, delirious					
CNS:	Verbalizing, anxious, confused and increasing obtunded				
HEENT:	Normal				
CVS:	Pulses bilaterally strong				
RESP:	Diffuse crackles bilaterally				
ABDO:	Soft, no distension, no pain on palpation				
GU:	Normal				
MSK:	Normal	SKIN:	Flushed and moist		

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: HR: 145/min BP: 150/80 RR: 32/min O ₂ SAT: 89% on 5L NP T: 40°C	Somnolent. When roused is confused and/delirious (will mumble - "I have to feed the cat, I need to wash the car; the plane is about to leave I have to go").	<u>Learner Actions</u> - ☐ History (including relevant to pregnancy) - ☐ Monitors - ☐ Px exam (including airway exam) - ☐ Replace NP with O ₂ mask - ☐ IV NS 500mL bolus - ☐ Blood work (Including blood cultures, VBG, TSH, LFTs) - ☐ Capillary glucose: 6.2 - ☐ CXR - ☐ ECG - ☐ IV Antibiotics	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 3 minutes → 2. Deterioration
2. Deterioration HR 150/min BP: 115/70 RR: 34/min O ₂ SAT: 93% (regardless of O ₂ used) GCS: 10	Patient is increasingly obtunded.	<u>Learner Actions</u> - ☐ Prepare for intubation - ☐ Discuss need to intubate with family members - ☐ Push-dose vasopressor at bedside - ☐ BiPap or PEEP valve to help pre-oxygenate - ☐ Intubate	<u>Modifiers</u> - No intubation by 8 min → drop O ₂ SAT to 85% - No discussion with husband → he questions what is happening <u>Triggers</u> - Intubation → 3. Labs Back
3. Labs Back HR 150/min Rhythm: a fib BP: 110/70 RR: 16/min (vented) O ₂ SAT: 93% (vented)	State begins by giving team lab results, including TSH. Patient intubated.	<u>Learner Actions</u> - ☐ Propranolol 0.5-1 mg IV slow push test first then q15 min to desired effect (or esmolol) - ☐ Recognize cardioversion unlikely to succeed or call cardiology for opinion - ☐ Hydrocortisone 300mg IV - ☐ Methimazole 20-25 mg PO/NG OR Propylthiouracil 600-1000 mg PO/NG - ☐ Initiate cooling - ☐ Lasix 40mg iv - ☐ Verbalize need for Lugol's solution (8 drops) or SSKI (5 drops) in 1 hour - ☐ Consult Endocrinology & ICU - ☐ Insert foley - ☐ Insert NG/OG to facilitate giving meds - ☐ Initiate post-intubation sedation	END CASE PRN

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

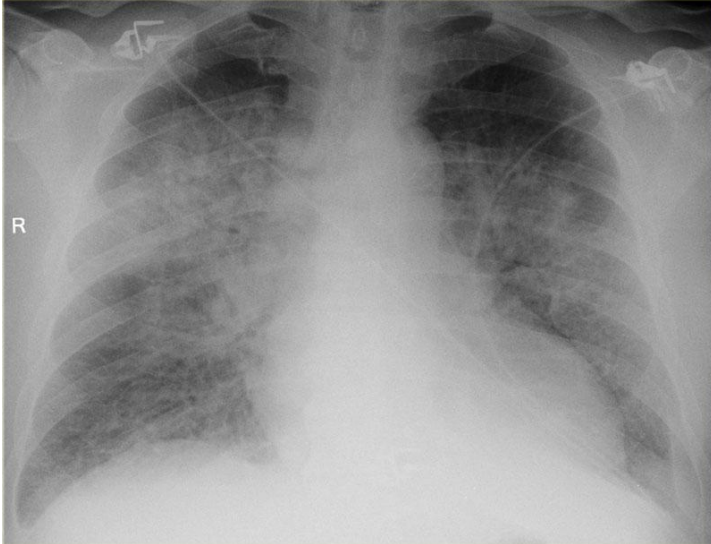
Laboratory Results						
Na: 144	K: 5	Cl: 109	HCO ₃ : 24	BUN: 11	Cr: 70	Glu: 6.9
Ca: 2.80		Mg: 0.94		TSH: 0.1		T3: pending T4: pending
ABG	pH: 7.33	PCO ₂ : 48	PO ₂ : 89	HCO ₃ : 24	Lactate: 3.2	
WBC 14		Hb 125		Plt 450		

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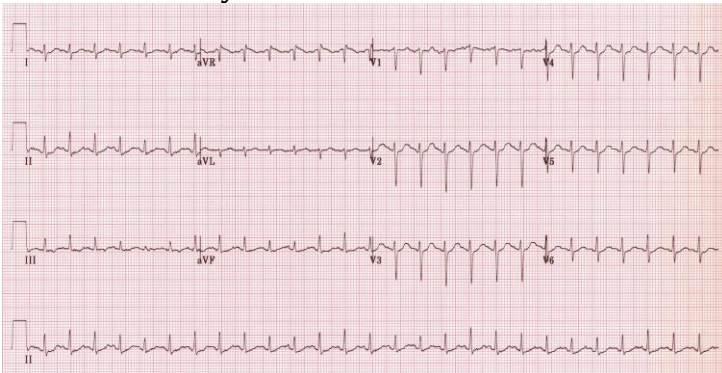
Images (ECGs, CXRs, etc.)

CXR: CHF



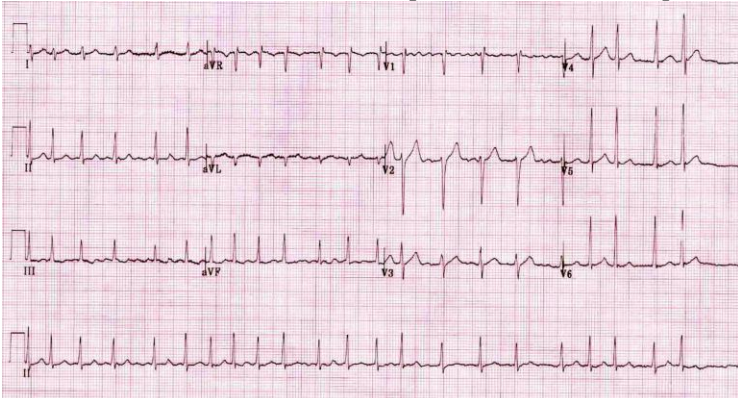
(CXR source: <http://www.radiologyassistant.nl/en/p4c132f36513d4/chest-x-ray-heart-failure.html>)

ECG: Sinus Tachycardia



(ECG source: <http://lifeinthefastlane.com/ecg-library/hyperthyroidism/>)

ECG: Atrial fibrillation with rapid ventricular response



(ECG source: <http://lifeinthefastlane.com/ecg-library/atrial-fibrillation/>)

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What is your differential diagnosis for fever with delirium? Sitting here thinking about it, is there anything you would have changed about your work-up of this patient, or are you happy with your orders? 2) What is the appropriate management of CHF in the context of a thyroid storm? 3) What is the management of thyroid storm? 4) How did it feel to discuss the need for intubation with the husband? 5) How did your team interact today? Do you feel like you were working through your differential together?	
Key Moments	
Identification of fever and broad work-up given altered mental status.	
Decision to intubate.	
Diagnosis and management of thyroid storm.	